

Greedy Problem's properties

- **Greedy Choice Property:** Choosing the best option at each phase can lead to a global (overall) optimal solution.
- **Optimal Substructure:** If an optimal solution to the complete problem contains the optimal solutions to the subproblems, the problem has an optimal substructure.

Problem list:

- 0/1 Knapsack Problem
- Coin Change (minimum number of coins)
- Finding maximum value from a tree

Can we find a greedy solution for these problems?

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NO

Coin Change

Number of coins 3: {1,3,4}, Target value: 6

Greedy Choice (choose the highest valued coins):
 $4 + 1 + 1$

Optimal Solution: $3+3$

0/1 Knapsack

Number of objects 3:

Weight: {1,2,3}

Value: {6,10,12}

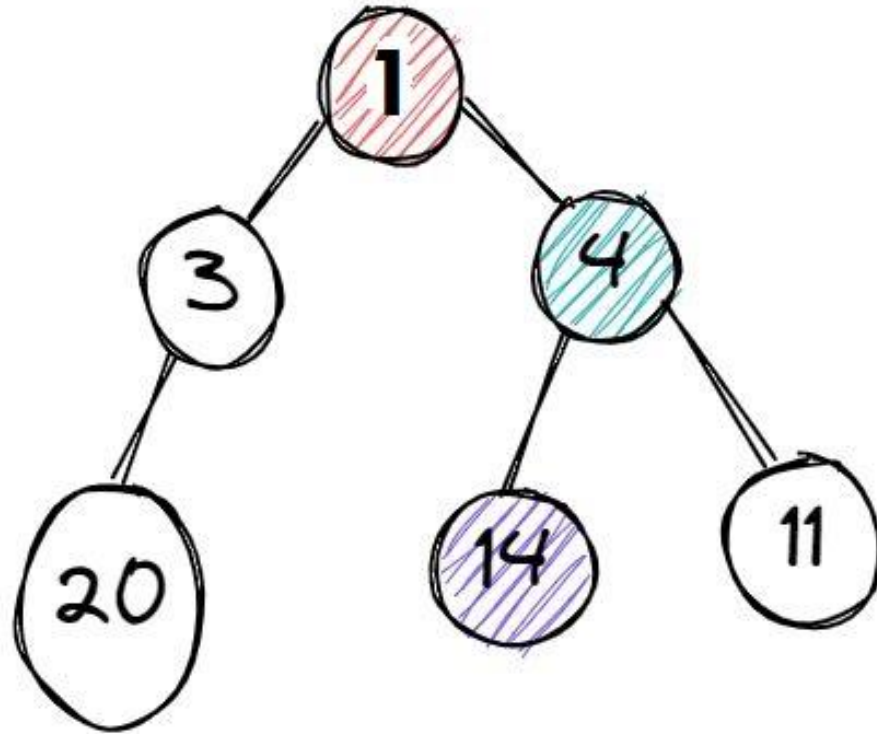
Value/Weight: {6, 5, 4}

Total Capacity: 5

Greedy Choice (choose the object with best value/weight ratio):
object1 + object2 (6+10): **16**

Optimal Solution: object2 + object 3 (10 + 12): **22**

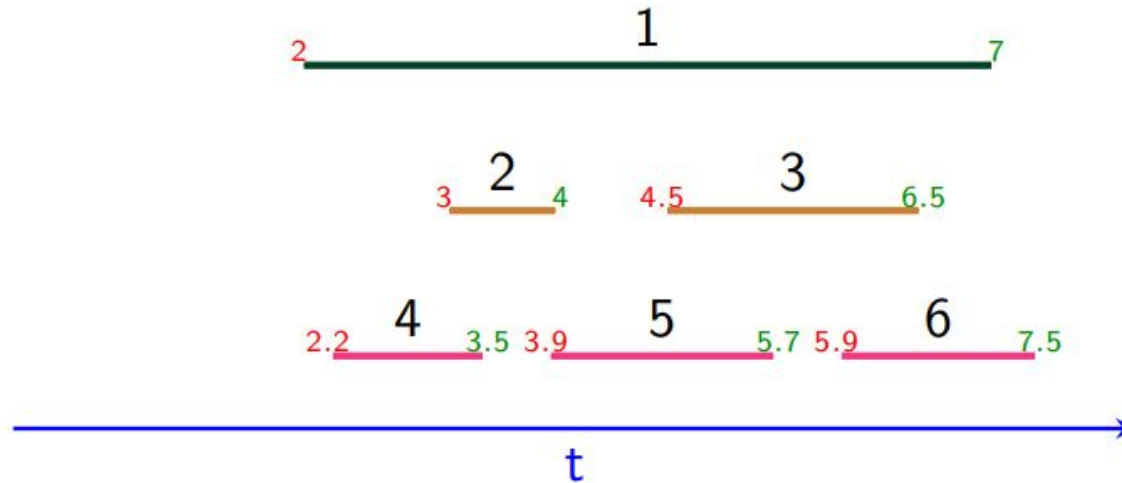
Maximum
Value from a
Tree



Problems for Greedy Algorithm

- Fractional knapsack problem
- Interval scheduling
- Job Scheduling
- Interval partitioning
- Huffman Encoding

Interval Scheduling (Activity selection)



- Single resource
- n requests
- Each with $s[i]$ start time and $f(i)$ finish time ($s[i] < f[i]$)
- Two requests i and j are compatible if they don't overlap
 - $f(i) \leq s(j)$, or
 - $f(j) \leq s(i)$
- **Goal:** Select a compatible subset of requests of maximum size

Greedy solution idea

- Sort the requests (based on what?)
- Select the requests from the sorted list, that does not overlap with the already selected requests

Interval Scheduling (Sorting Options)

- Earlier start time
- Smallest request
- Minimum overlap (fewest conflicts)
- Earlier finish time

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 - (1,6) (2,3) (4,5)
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 - (5,8) (1,6) (7,15)
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- Earlier start time
 - (1,6) (2,3) (4,5)
- Smallest request
 - (5,8) (1,6) (7,15)
- Minimum overlap (fewest conflicts)
 - (1,3) (4,6) (7,9) (10,12) (5,8) (2,5) (2,5) (2,5) (8,11) (8,11) (8,11)

Counter examples



counterexample for earliest start time



counterexample for shortest interval



counterexample for fewest conflicts

Interval Scheduling (Activity Selection)

Sort all activities by their end times (ascending).

Initialize `last_end_time = 0`

For each activity (start, end) in sorted list:

 If `start >= last_end_time`:

 Select the activity

 Update `last_end_time = end`

Proof of correctness

Let G , be the set of selected intervals by greedy algorithm
and

O , be the set of optimal intervals

If $G = O$, then G is already optimal. Nothing to prove.

Proof of correctness

Assume, $O \neq G$

Let,

$$G = \{g_1, g_2, \dots, g_m\}$$

$$O = \{r_1, r_2, \dots, r_n\}$$

As the Greedy solution can not have more elements than optimal solution, so: **$n \geq m$**

Proof of correctness

Assume k^{th} element is the first mismatched interval between O and G

$$G = \{g_1, g_2, g_{k-1}, g_k, \dots, g_m\}$$

$$O = \{g_1, g_2, g_{k-1}, r_k, \dots, r_n\}$$

Can we replace r_k with g_k without contradiction and O still remain Optimal?

Proof of correctness

As both of g_k and r_k is compatible with g_{k-1} , so different start time does not create any conflict.

The greedy choice guarantee that, the finish time of g_k can not be greater than finish time r_k ($f_{g_k} \leq f_{r_k}$). As r_k is not currently interfering with the next element (r_{k+1}) in O , so g_k will not also interfere with it.

So, we can replace r_k with g_k without creating conflict with r_{k+1}

After replacement:

$$G = \{g_1, g_2, g_{k-1}, g_k, \dots, g_m\}$$

$$O = \{g_1, g_2, g_{k-1}, g_k, \dots, r_n\}$$

Proof of correctness

By repeating this process, we can replace all g_i to the set O

$$G = \{g_1, g_2, g_{k-1}, g_k, \dots, g_m\}$$

$$O = \{g_1, g_2, g_{k-1}, g_k, \dots, g_m, \dots, r_n\}$$

But this can't be true, because if there is no conflict, then Greedy algorithm keeps adding intervals. So, if there are non-conflicting elements in O after g_m , that should be picked by greedy algorithm and would be part of G .

So $|G| = |O|$, and we can convert G to O , thus proving the optimality of G

Interval Partitioning

Input A set of lectures, with start and end times

Goal Find the minimum number of classrooms, needed to schedule all the lectures such two lectures do not occur at the same time in the same room.

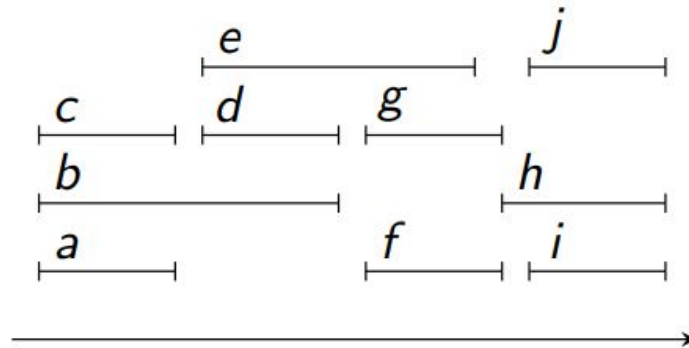


Figure: A schedule requiring 4 classrooms

Solution Idea

- Apply interval scheduling, and assign a room for the selected lectures
- Apply interval scheduling for the remaining lectures and add a new room. (do this until there are no lectures remaining)
- Will it work?

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- Apply interval scheduling, and assign a room for the selected lectures
- Apply interval scheduling for the remaining lectures and add a new room. (do this until there are no lectures remaining)
- Will it work? (NO)

Counter example

(3,7) (5,9) (8,11) (12,13) (10,14) (16,18) (13,20)

- (3,7) (8,11) (12,13) (16,18)
- (5,9) (10,14)
- (13,20)

Greedy Solution

Sort all lectures by their starting time

set: $r = 0$

for $j=1$ to n

 if (there is a room k , which is compatible with j)

 assign j to room k

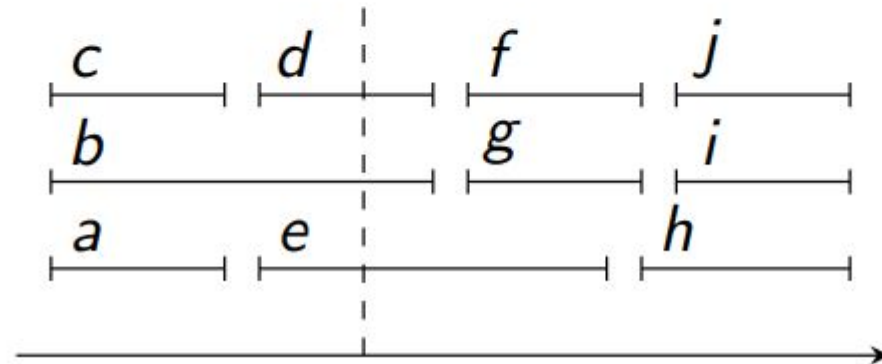
 else

 allocate a new room $r = r+1$

 assign j to $r+1$

Proof of correctness

- For a set of lectures R , k are said to be **in conflict** if there is some time t such that there are k lectures going on at time t .
- The **depth** of a set of lectures R is the maximum number of lectures in conflict at any time.



Proof of correctness

For any set R of lectures, the number of class-rooms required is at least the depth of R .

Proof.

All lectures that are in conflict must be scheduled in different rooms. □

Proof of correctness

Let d be the depth of the set of lectures R . The number of class-rooms used by the greedy algorithm is d .

Proof.

- Suppose the greedy algorithm uses more than d rooms. Let j be the first lecture that is scheduled in room $d + 1$.
- Since we process lectures according to start times, there are d lectures that start (at or) before j and which are in conflict with j .
- Thus, *at the start time of j* , there are at least $d + 1$ lectures in conflict, which contradicts the fact that the depth is d . $\square$

Job sequencing

- Given an **array of jobs** where every job has:
 - A deadline
 - Associated profit if the job is finished before the deadline.
- It is also given that every job takes a single unit of time, so the minimum possible deadline for any job is 1.

Target is to **maximize the total profit** if only one job can be scheduled at a time.

Input: Four Jobs with following deadlines and profits

JobID	Deadline	Profit
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a	4	20
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b	1	10
---	---	----

c	1	40
---	---	----

d	1	30
---	---	----

Output: Following is maximum profit sequence of jobs

c, a

Input: Five Jobs with following deadlines and profits

JobID	Deadline	Profit
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a	2	100
---	---	-----

b	1	19
---	---	----

c	2	27
---	---	----

d	1	25
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e	3	15
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Output: Following is maximum profit sequence of jobs

c, a, e

Greedy Solution of Job sequencing

Idea is to select the most profitable job, but complete it as delayed as possible

- Sort all jobs in decreasing order of profit.
- Iterate on jobs in decreasing order of profit. For each job , do the following :
 - Find a time slot i , such that slot is empty and $i < \text{deadline}$ and i is greatest. Put the job in this slot and mark this slot filled.
 - If no such i exists, then ignore the job.

Proof of correctness

If Greedy and Optimal aren't identical, there must be a first time slot where they disagree. Let's look at that specific slot:

- Greedy put Job A in this slot.
- Optimal put Job B in this slot (or left it empty).